

5 (a) (i) $I_1 + I_3 = I_2$ A1 [1]

(ii) $E_1 = I_2 \frac{R_2}{2} + I_1 \frac{R_2}{2} + I_1 R_1 + I_1 r_1$ A1 [1]

(iii) $E_1 - E_2$ B1
 $= -I_3 r_2 + I_1 (R_1 + r_1 + R_2 / 2)$ B1 [2]

(b) p.d. across BJ of wire changes / resistance of BJ changes B1
 there is a difference in p.d across wire and p.d. across cell E_2 B1 [2]

6 (a) waves overlap B1